
BY 531 : Quantitative Biology and Data Analytics

Course Code : BY 531

Course Name : Quantitative Biology and Data Analytics

L-T-P-C : 3-0-0-3

Intended for : Core for M.Tech Biotechnology and elective for B.Tech, M.Sc/M.Tech and PhD candidates

Prerequisite : IC136 - Understanding Biotechnology & its Applications or Consent of faculty member

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Module 1:** Introduction to statistics and probability theory (distributions, mean, standard deviations, variance, etc.), basic principles of statistical inference, statistical models, linear models, basic bayesian methods, confidence interval estimation and hypothesis testing, including P-values, significance level, power, sample size, and two types of errors. (4 hours)
- **Module 2:** Linear algebra and matrices, inference for high dimensional data, distance and dimension reduction, PCA, t-SNE, etc., machine learning, and batch effects. (8 hours)
- **Module 3:** Data visualization, exploratory data analysis, and robust summaries. (4 hours)
- **Module 4:** Mathematical models for biological processes, basic calculus, logistic regression, bifurcation and steady state analysis in population growth models, epidemic models, and predator-prey models. (6 hours)
- **Module 5:** Image processing in MATLAB: types of image - binary, grayscale, & color; image thresholding and segmentation; particle tracking, and z-stack projections. (8 hours)

Textbooks:

1. Brian Munsky et. al., **Quantitative Biology: Theory, Computational Methods, and Models**.
2. Whitlock, Michael C.; Schluter, Dolph., **The Analysis of Biological Data**, 2nd Edition, Freeman, W. H. & Company, 2014

References:

1. Data Science for Biology, Emily Ren et. al., Online materials, etc.